

30. 8-Bit Timer (TMRb)

This MCU has two units (unit 0, unit 1) of an on-chip 8-bit timer (TMR) module that comprise two 8-bit counter channels, totaling four channels. The 8-bit timer module can be used to count external events and also be used as a multi-function timer in a variety of applications, such as generation of counter reset signal, interrupt requests, and pulse output with a desired duty cycle using a compare-match signal with two registers.

Unit 0 and unit 1 have the same functions, and can generate a base clock for the SCI.

In this section, “PCLK” is used to refer to PCLKB.

30.1 Overview

Table 30.1 lists the specifications of the TMR. Table 30.2 lists the TMR functions.

Figure 30.1 shows a block diagram of the 8-bit timer module (unit 0), and Figure 30.2 shows that of the 8-bit timer module (unit 1).

Table 30.1 Specifications of TMR

Item	Description
Count clock	- Internal clock: PCLK/1, PCLK/2, PCLK/8, PCLK/32, PCLK/64, PCLK/1024, PCLK/8192 - External clock: external count clock
Number of channels	(8 bits × 2 channels) × 2 units
Compare match	8-bit mode (compare match A, compare match B) 16-bit mode (compare match A, compare match B)
Counter clear	Selected by compare match A or B, or an external counter reset signal.
Timer output	Output pulses with a desired duty cycle or PWM output
Cascading of two channels	16-bit count mode 16-bit timer using TMR0 for the upper 8 bits and TMR1 for the lower 8 bits (TMR2 for the upper 8 bits and TMR3 for the lower 8 bits) - Compare match count mode TMR1 can be used to count TMR0 compare matches (TMR3 can be used to count TMR2 compare matches).
Interrupt sources	Compare match A, compare match B, and overflow
Event link function (Output)	Compare match A, compare match B, and overflow (TMR0 to TMR3)
Event link function (Input)	One of the following three operations proceeds in response to an event reception: (1) Counting start operation (TMR0 to TMR3) (2) Event counting operation (TMR0 to TMR3) (3) Counting restart operation (TMR0 to TMR3)
DTC activation	DTC can be activated by compare match A interrupts or compare match B interrupts.
A/D conversion start trigger of the A/D converter	Compare match A of TMR0 and TMR2
Capable of generating base clock for SCI	Generates base clock for SCI.*1
Low power consumption function	Each unit can be placed in a module stop state

Note 1. For details, refer to section 41, Serial Communications Interface (SCIj, SCli, SCIh).

Table 30.2 TMR Functions

Item		Unit 0			Unit 1		
Counter mode		8 Bits		16 Bits	8 Bits		16 Bits
Channel		TMR0	TMR1	TMR0 + TMR1	TMR2	TMR3	TMR2 + TMR3
Count clock		PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi0	PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi1	PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi1	PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi2	PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi3	PCLK/1 PCLK/2 PCLK/8 PCLK/32 PCLK/64 PCLK/1024 PCLK/8192 TMCi3
Counter clear		TMR0.TCORA TMR0.TCORB TMRi0	TMR1.TCORA TMR1.TCORB TMRi1	TMR0.TCORA + TMR1.TCORA TMR0.TCORB + TMR1.TCORB TMRi0	TMR2.TCORA TMR2.TCORB TMRi2	TMR3.TCORA TMR3.TCORB TMRi3	TMR2.TCORA + TMR3.TCORA TMR2.TCORB + TMR3.TCORB TMRi2
Compare match	Compare match A	✓	✓	✓	✓	✓	✓
	Compare match B	✓	✓	✓	✓	✓	✓
Timer output	Low output	✓	✓	✓	✓	✓	✓
	High output	✓	✓	✓	✓	✓	✓
	Toggle output	✓	✓	✓	✓	✓	✓
DTC activation	Compare match A	✓	✓	✓	✓	✓	✓
	Compare match B	✓	✓	✓	✓	✓	✓
	TCNT overflow	—	—	—	—	—	—
Interrupt	Compare match A	CMIA0	CMIA1	CMIA0	CMIA2	CMIA3	CMIA2
	Compare match B	CMIB0	CMIB1	CMIB0	CMIB2	CMIB3	CMIB2
	TCNT overflow	OVI0	OVI1	OVI0	OVI2	OVI3	OVI2
Cascaded connection		TMR1 overflow	TMR0 compare match A	—	TMR3 overflow	TMR2 compare match A	—
A/D conversion start trigger of the A/D converter*1		✓	—	✓	✓	—	✓
SCI base clock generation*2		✓		—	✓		—
ELC output event	Compare match A	✓	✓	✓	✓	✓	✓
	Compare match B	✓	✓	✓	✓	✓	✓
	TCNT overflow	✓	✓	✓	✓	✓	✓
ELC input event	Counting start	✓	✓	—	✓	✓	—
	Event counting	✓	✓	—	✓	✓	—
	Counting restart	✓	✓	—	✓	✓	—
Module stop setting*3		MSTPCRA.MSTPA5 bit (unit 0), MSTPCRA.MSTPA4 bit (unit 1)					

✓: Possible

—: Impossible

Note 1. For details, refer to section 56, 12-Bit A/D Converter (S12ADFa).

Note 2. For details, refer to section 41, Serial Communications Interface (SCIj, SCli, SCih).

Note 3. For details, refer to section 11, Low Power Consumption.

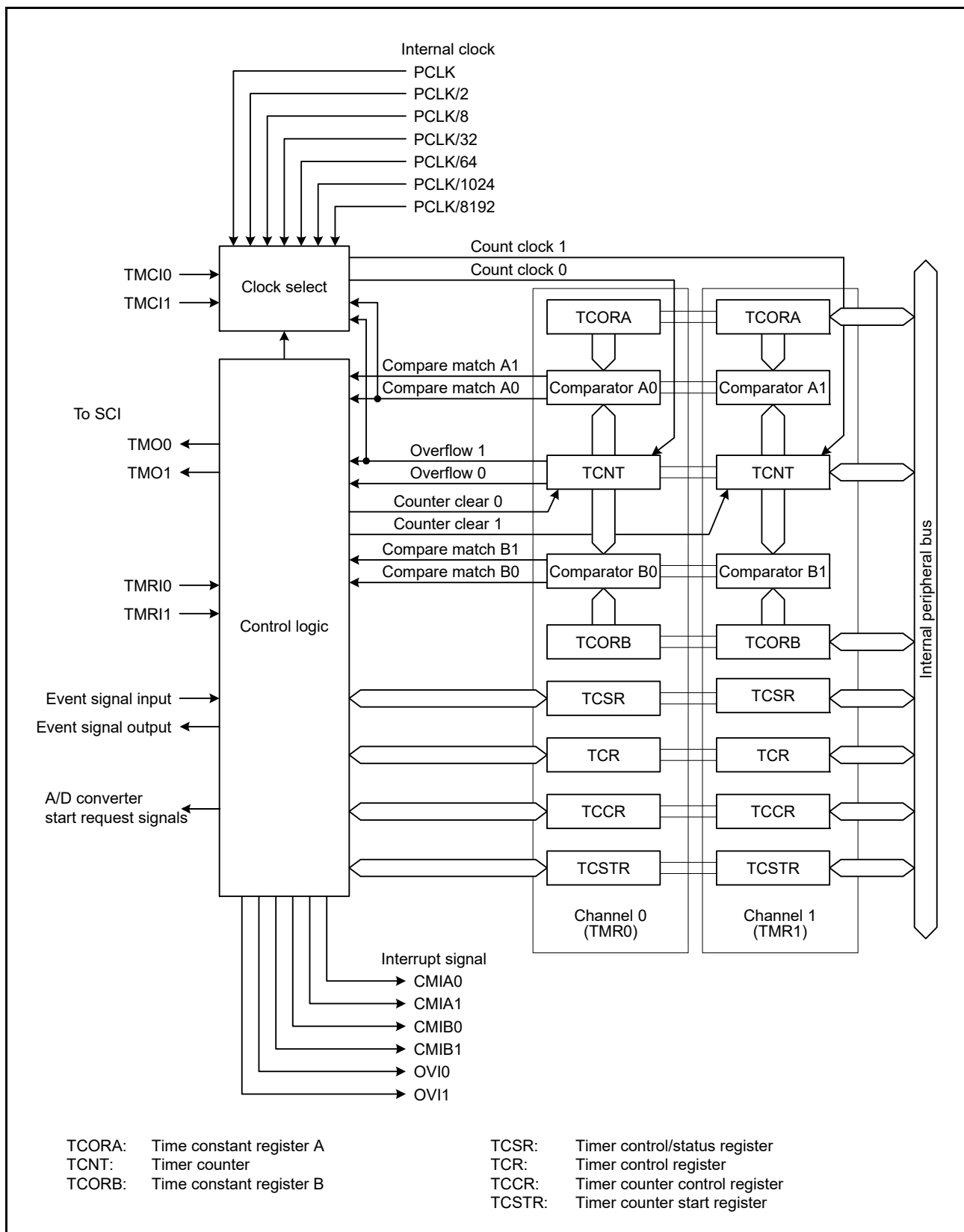


Figure 30.1 Block Diagram of TMR (Unit 0)

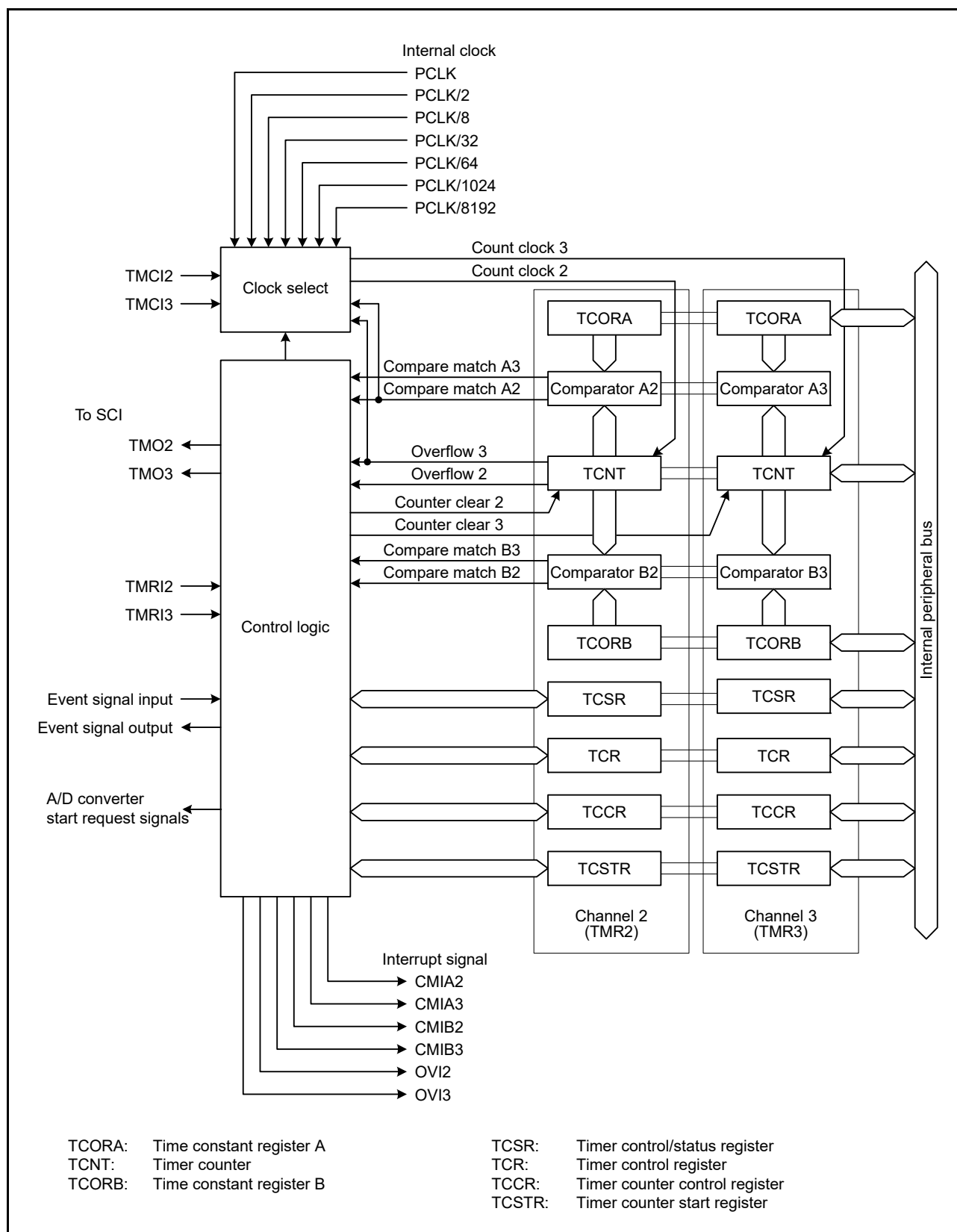


Figure 30.2 Block Diagram of TMR (Unit 1)

Table 30.3 lists the I/O pins of the TMR.

Table 30.3 Pin Configuration of TMR

Unit	Channel	Pin Name	I/O	Description
0	TMR0	TMO0	Output	Outputs compare match
		TMCi0	Input	Inputs external count clock
		TMRi0	Input	Inputs external counter reset
	TMR1	TMO1	Output	Outputs compare match
		TMCi1	Input	Inputs external count clock
		TMRi1	Input	Inputs external counter reset
1	TMR2	TMO2	Output	Outputs compare match
		TMCi2	Input	Inputs external count clock
		TMRi2	Input	Inputs external counter reset
	TMR3	TMO3	Output	Outputs compare match
		TMCi3	Input	Inputs external count clock
		TMRi3	Input	Inputs external counter reset

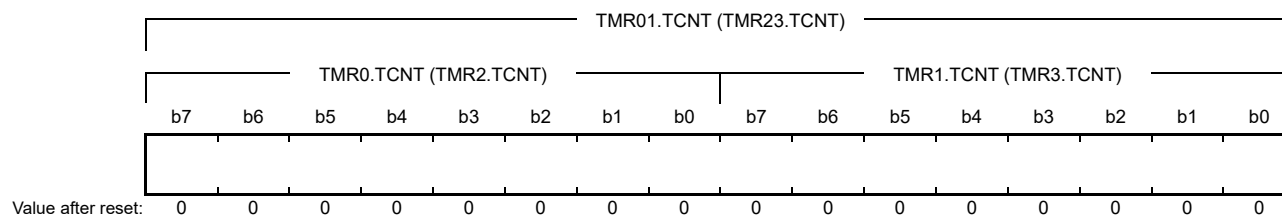
30.2 Register Descriptions

Table 30.4 Register Allocation for 16-Bit Access

Address	Register	Upper 8 Bits	Lower 8 Bits
0008 8208h	TMR01.TCNT	TMR0.TCNT	TMR1.TCNT
0008 8204h	TMR01.TCORA	TMR0.TCORA	TMR1.TCORA
0008 8206h	TMR01.TCORB	TMR0.TCORB	TMR1.TCORB
0008 820Ah	TMR01.TCCR	TMR0.TCCR	TMR1.TCCR
0008 8218h	TMR23.TCNT	TMR2.TCNT	TMR3.TCNT
0008 8214h	TMR23.TCORA	TMR2.TCORA	TMR3.TCORA
0008 8216h	TMR23.TCORB	TMR2.TCORB	TMR3.TCORB
0008 821Ah	TMR23.TCCR	TMR2.TCCR	TMR3.TCCR

30.2.1 Timer Counter (TCNT)

Address(es): TMR0.TCNT 0008 8208h, TMR1.TCNT 0008 8209h, TMR2.TCNT 0008 8218h, TMR3.TCNT 0008 8219h,
TMR01.TCNT 0008 8208h, TMR23.TCNT 0008 8218h



TCNT is an 8-bit readable/writable up-counter.

TMR0.TCNT and TMR1.TCNT (TMR2.TCNT and TMR3.TCNT) comprise a single 16-bit counter (TMR01.TCNT, TMR23.TCNT) so they can be accessed together in 16-bit units.

The TCCR.CSS[1:0] and CKS[2:0] bits are used to select a count clock.

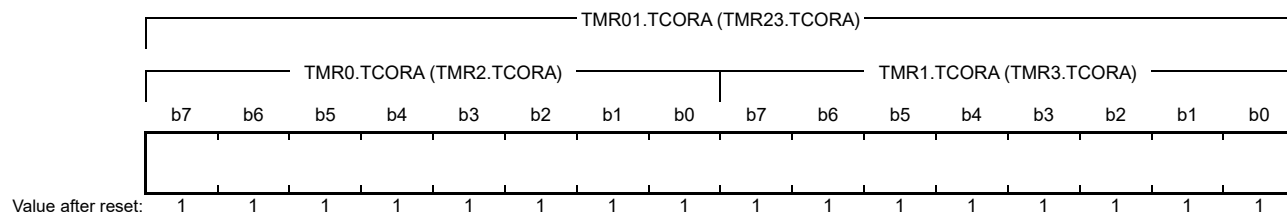
TCNT can be cleared by an external counter reset signal, compare match A, or compare match B. Which compare match to be used for clearing is selected by the TCR.CCLR[1:0] bits.

When TCNT overflows (its value changes from FFh to 00h), an overflow interrupt is output provided the interrupt request is enabled by the TCR.OVIE bit.

For details on the corresponding interrupt vector number, refer to section 15, Interrupt Controller (ICUD), and Table 30.6, TMR Interrupt Sources.

30.2.2 Time Constant Register A (TCORA)

Address(es): TMR0.TCORA 0008 8204h, TMR1.TCORA 0008 8205h, TMR2.TCORA 0008 8214h, TMR3.TCORA 0008 8215h,
TMR01.TCORA 0008 8204h, TMR23.TCORA 0008 8214h



TCORA is an 8-bit readable/writable register.

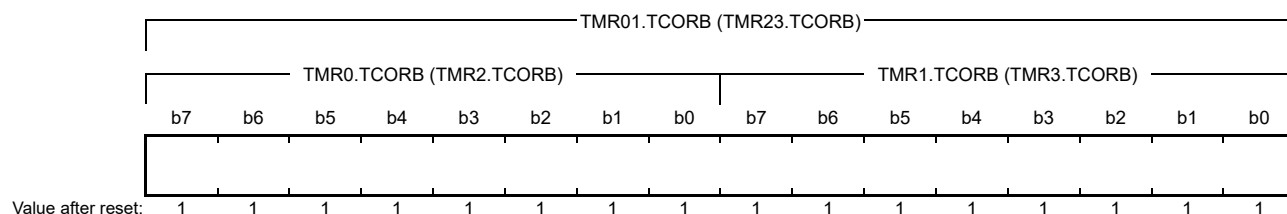
TMR0.TCORA and TMR1.TCORA (TMR2.TCORA and TMR3.TCORA) comprise a single 16-bit register (TMR01.TCORA, TMR23.TCORA) so they can be accessed together in 16-bit units.

The value in TCORA is continually compared with the value in TCNT. When a match is detected, the corresponding compare match A is generated, and a compare match A interrupt is output provided the interrupt request is enabled by the TCR.CMIEA bit.

However, comparison is not performed during writing to TCORA. The timer output from the TMO pin can be freely controlled by this compare match A and the settings of the TCSR.OSA[1:0] bits.

30.2.3 Time Constant Register B (TCORB)

Address(es): TMR0.TCORB 0008 8206h, TMR1.TCORB 0008 8207h, TMR2.TCORB 0008 8216h, TMR3.TCORB 0008 8217h,
TMR01.TCORB 0008 8206h, TMR23.TCORB 0008 8216h



TCORB is an 8-bit readable/writable register.

TMR0.TCORB and TMR1.TCORB (TMR2.TCORB and TMR3.TCORB) comprise a single 16-bit register (TMR01.TCORB, TMR23.TCORB) so they can be accessed together in 16-bit units.

The value in TCORB is continually compared with the value in TCNT. When a match is detected, the corresponding compare match B is generated, and a compare match B interrupt is output provided the interrupt request is enabled by the TCR.CMIEB bit.

However, comparison is not performed during writing to TCORB. The timer output from the TMO pin can be freely controlled by this compare match B and the settings of the TCSR.OSB[1:0] bits.

30.2.4 Timer Control Register (TCR)

Address(es): TMR0.TCR 0008 8200h, TMR1.TCR 0008 8201h, TMR2.TCR 0008 8210h, TMR3.TCR 0008 8211h

b7	b6	b5	b4	b3	b2	b1	b0
CMIEB	CMIEA	OVIE	CCLR[1:0]	—	—	—	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4, b3	CCLR[1:0]	Counter Clear	b4 b3 0 0: Clearing is disabled 0 1: Cleared by compare match A 1 0: Cleared by compare match B 1 1: Cleared by the external counter reset signal*1 (Select edge or level by the TMRIS bit in TCCR.)	R/W
b5	OVIE	Overflow Interrupt Enable	0: Overflow interrupt requests (OVIn) are disabled 1: Overflow interrupt requests (OVIn) are enabled	R/W
b6	CMIEA	Compare Match A Interrupt Enable	0: Compare match A interrupt requests (CMIAIn) are disabled 1: Compare match A interrupt requests (CMIAIn) are enabled	R/W
b7	CMIEB	Compare Match B Interrupt Enable	0: Compare match B interrupt requests (CMIBIn) are disabled 1: Compare match B interrupt requests (CMIBIn) are enabled	R/W

Note 1. To use an external counter reset signal, set the corresponding pin function. For details, refer to section 22, I/O Ports and section 23, Multi-Function Pin Controller (MPC).

CCLR[1:0] Bits (Counter Clear)

Select the condition by which TCNT is cleared.

OVIE Bit (Overflow Interrupt Enable)

Selects whether overflow interrupt requests (OVIn) issued by TCNT are enabled or disabled.

CMIEA Bit (Compare Match A Interrupt Enable)

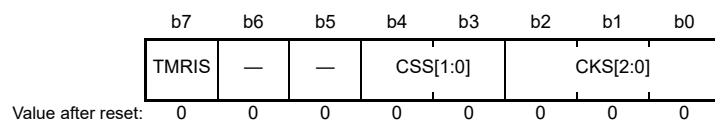
Selects whether compare match A interrupt requests (CMIAIn) that are issued when the value of TCORA corresponds to that of TCNT are enabled or disabled.

CMIEB Bit (Compare Match B Interrupt Enable)

Selects whether compare match B interrupt requests (CMIBIn) that are issued when the value of TCORB corresponds to that of TCNT are enabled or disabled.

30.2.5 Timer Counter Control Register (TCCR)

Address(es): TMR0.TCCR 0008 820Ah, TMR1.TCCR 0008 820Bh, TMR2.TCCR 0008 821Ah, TMR3.TCCR 0008 821Bh,
TMR01.TCCR 0008 820Ah, TMR23.TCCR 0008 821Ah



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	CKS[2:0]	Clock Select* ¹	See Table 30.5.	R/W
b4, b3	CSS[1:0]	Clock Source Select	See Table 30.5.	R/W
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	TMRIS	Timer Reset Detection Condition Select	0: Cleared at rising edge of the external counter reset signal 1: Cleared when the external counter reset signal is high	R/W

Note 1. To use an external count clock, set the corresponding pin function. For details, refer to section 22, I/O Ports and section 23, Multi-Function Pin Controller (MPC).

TCCR register is a 8-bit register used to configure the basic operation of the counter. Two TCCR registers can be accessed simultaneously by accessing the address of the even channel TCCR register in 16-bit units.

CKS[2:0] Bits (Clock Select)

CSS[1:0] Bits (Clock Source Select)

The CKS[2:0] and CSS[1:0] bits select a clock. For details, see Table 30.5.

TMRIS Bit (Timer Reset Detection Condition Select)

This bit is enabled when the TCR.CCLR[1:0] bits are 11b (cleared by external counter reset signal) and selects the condition for detecting counter reset (level or edge).

Table 30.5 Clock Input to TCNT and Count Condition

Channel	TCCR Register					Description
	CSS[1:0]		CKS[2:0]			
	b4	b3	b2	b1	b0	
TMR0 (TMR2)	0	0	—	0	0	Clock input prohibited
					1	Uses external count clock. Counts at rising edge*1.
				1	0	Uses external count clock. Counts at falling edge*1.
					1	Uses external count clock. Counts at both rising and falling edges*1.
	0	1	0	0	0	Uses internal clock. Counts at PCLK.
					1	Uses internal clock. Counts at PCLK/2.
				1	0	Uses internal clock. Counts at PCLK/8.
					1	Uses internal clock. Counts at PCLK/32.
			1	0	0	Uses internal clock. Counts at PCLK/64.
					1	Uses internal clock. Counts at PCLK/1024.
				1	0	Uses internal clock. Counts at PCLK/8192.
					1	Clock input prohibited
	1	0	—	—	—	Setting prohibited
	1	1	—	—	—	Counts at TMR1.TCNT (TMR3.TCNT) overflow signal*2.
TMR1 (TMR3)	0	0	—	0	0	Clock input prohibited
					1	Uses external count clock. Counts at rising edge*1.
				1	0	Uses external count clock. Counts at falling edge*1.
					1	Uses external count clock. Counts at both rising and falling edges*1.
	0	1	0	0	0	Uses internal clock. Counts at PCLK.
					1	Uses internal clock. Counts at PCLK/2.
				1	0	Uses internal clock. Counts at PCLK/8.
					1	Uses internal clock. Counts at PCLK/32.
			1	0	0	Uses internal clock. Counts at PCLK/64.
					1	Uses internal clock. Counts at PCLK/1024.
				1	0	Uses internal clock. Counts at PCLK/8192.
					1	Clock input prohibited
	1	0	—	—	—	Setting prohibited
	1	1	—	—	—	Counts at TMR0.TCNT (TMR2.TCNT) compare match A*2.

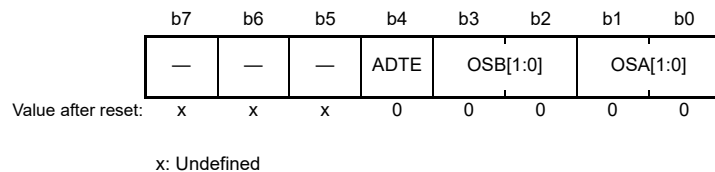
Note 1. To use an external count clock, set the corresponding pin function. For details, refer to section 22, I/O Ports and section 23, Multi-Function Pin Controller (MPC).

Note 2. If the clock input of TMR0 (TMR2) is the overflow signal of the TMR1.TCNT (TMR3.TCNT) counter and that of TMR1 (TMR3) is the compare match signal of the TMR0.TCNT (TMR2.TCNT) counter, no TCNT count clock is generated. Do not use this setting.

30.2.6 Timer Control/Status Register (TCSR)

- TMR0.TCSR, TMR2.TCSR

Address(es): TMR0.TCSR 0008 8202h, TMR2.TCSR 0008 8212h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	OSA[1:0]	Output Select A* ¹	b1 b0 0 0: No change 0 1: Low is output 1 0: High is output 1 1: Output is inverted (toggle output)	R/W
b3, b2	OSB[1:0]	Output Select B* ¹	b3 b2 0 0: No change 0 1: Low is output 1 0: High is output 1 1: Output is inverted (toggle output)	R/W
b4	ADTE	A/D Trigger Enable	0: A/D conversion start request in response to compare match A is disabled. 1: A/D conversion start request in response to compare match A is enabled.	R/W
b7 to b5	—	Reserved	These bits are read as an undefined value. The write value should be 1.	R/W

Note 1. When all OSA[1:0] and OSB[1:0] bits are 0, the output enable signal corresponding to the TMO_n pin is negated and a request for high-impedance output is issued to the I/O port. Timer output pin is driven low until the first compare-match occurs after a reset when at least one of the OSA[1:0] and OSB[1:0] bits is 1.

OSA[1:0] Bits (Output Select A)

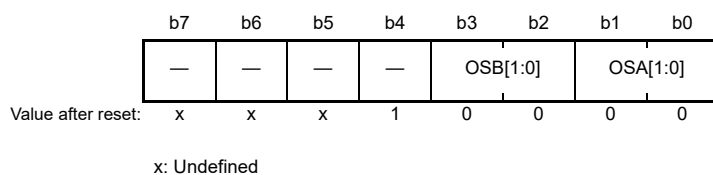
These bits select a method of TMO_n pin output when compare match A of TCORA and TCNT occurs.

OSB[1:0] Bits (Output Select B)

These bits select a method of TMO_n pin output when compare match B of TCORB and TCNT occurs.

- TMR1.TCSR, TMR3.TCSR

Address(es): TMR1.TCSR 0008 8203h, TMR3.TCSR 0008 8213h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	OSA[1:0]	Output Select A*1	b1 b0 0 0: No change 0 1: Low is output 1 0: High is output 1 1: Output is inverted (toggle output)	R/W
b3, b2	OSB[1:0]	Output Select B*1	b3 b2 0 0: No change 0 1: Low is output 1 0: High is output 1 1: Output is inverted (toggle output)	R/W
b4	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b7 to b5	—	Reserved	These bits are read as an undefined value. The write value should be 1.	R/W

Note 1. When all OSA[1:0] and OSB[1:0] bits are 0, the output enable signal corresponding to the TMO pin is negated and a request for high-impedance output is issued to the I/O port. Timer output pin is driven low until the first compare match occurs after a reset when at least one of the OSA[1:0] and OSB[1:0] bits is 1.

OSA[1:0] Bits (Output Select A)

These bits select a method of TMO pin output when compare match A of TCORA and TCNT occurs.

OSB[1:0] Bits (Output Select B)

These bits select a method of TMO pin output when compare match B of TCORB and TCNT occurs.

30.2.7 Timer Counter Start Register (TCSTR)

Address(es): TMR0.TCSTR 0008 820Ch, TMR1.TCSTR 0008 820Dh, TMR2.TCSTR 0008 821Ch, TMR3.TCSTR 0008 821Dh

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	TCS
Value after reset:	x	x	x	x	x	x	x	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	TCS	Timer Counter Status	0: Count stopped state in response to ELC. 1: Count start state in response to ELC.	R/W
b7 to b1	—	Reserved	These bits are read as an undefined value. The write value should be 0.	R/W

TCS Bit (Timer Counter Status)

The TCS bit is used to check the state of the timer count in response to ELC.

When this bit is read as 1, it shows the timer start state in response to ELC. When this bit is read as 0, it shows the timer stopped state in response to ELC.

This bit is cleared by writing 0. Do not write 1 to this bit.

The TCS bit is valid only when the count start operation is selected by the ELOPD register of the event controller (ELC). For details, refer to section 30.7, Link Operation by ELC, or section 21, Event Link Controller (ELC).

30.3 Operation

30.3.1 Pulse Output

Figure 30.3 shows an example of the 8-bit timer being used to generate a pulse output with a desired duty cycle.

1. Set the TCR.CCLR[1:0] bits to 01b (cleared by compare match A) so that TCNT is cleared at a compare match of TCORA.
2. Set the TCSR.OSA[1:0] bits to 10b (high is output) and TCSR.OSB[1:0] bits to 01b (low is output), causing the output to change to high at a compare match of TCORA and to low at a compare match of TCORB.

With these settings, the 8-bit timer provides pulses output at a cycle determined by TCORA with a pulse width determined by TCORB. No software intervention is required.

The timer output pin is low after the TCSR.OSA[1:0] or TCSR.OSB[1:0] bits are set until the first compare match occurs after a reset.

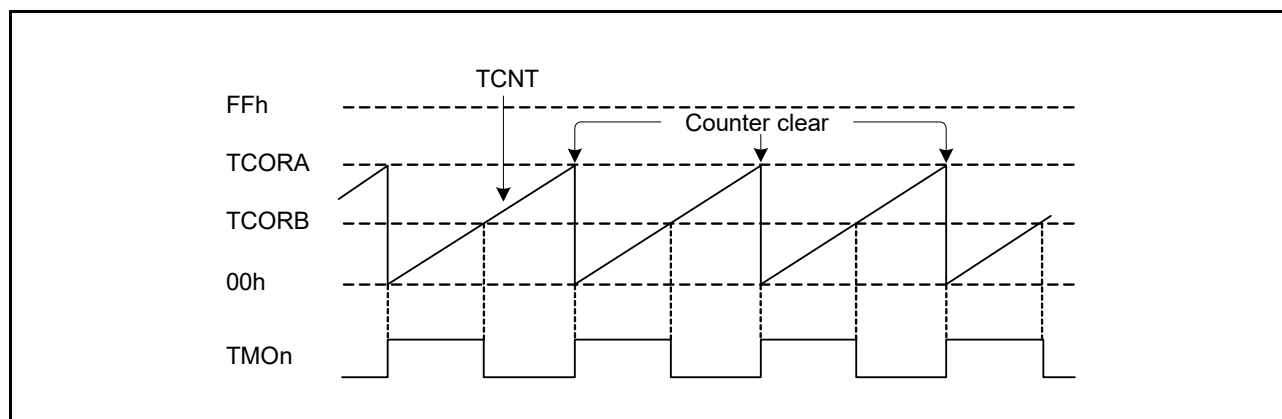


Figure 30.3 Example of Pulse Output (n = 0 to 3)

30.3.2 External Counter Reset Input

Figure 30.4 shows an example of the 8-bit timer being used to generate a pulse which is output after a desired delay time from a TMRIn input.

1. Set the TCR.CCLR[1:0] bits to 11b (cleared by external counter reset signal) and set the TMRIS bit in TCCR to 1 (cleared when the external counter reset signal is high) so that TCNT is cleared at the high level input of the TMRIn signal.
2. Set the TCSR.OSA[1:0] bits to 10b (high output) and the TCSR.OSB[1:0] bits to 01b (low output), causing the output to change to high at a compare match of TCORA and to low at a compare match of TCORB.

With these settings, the 8-bit timer provides pulses output at a desired delay time from a TMRIn input determined by TCORA and with a pulse width determined by TCORB and TCORA.

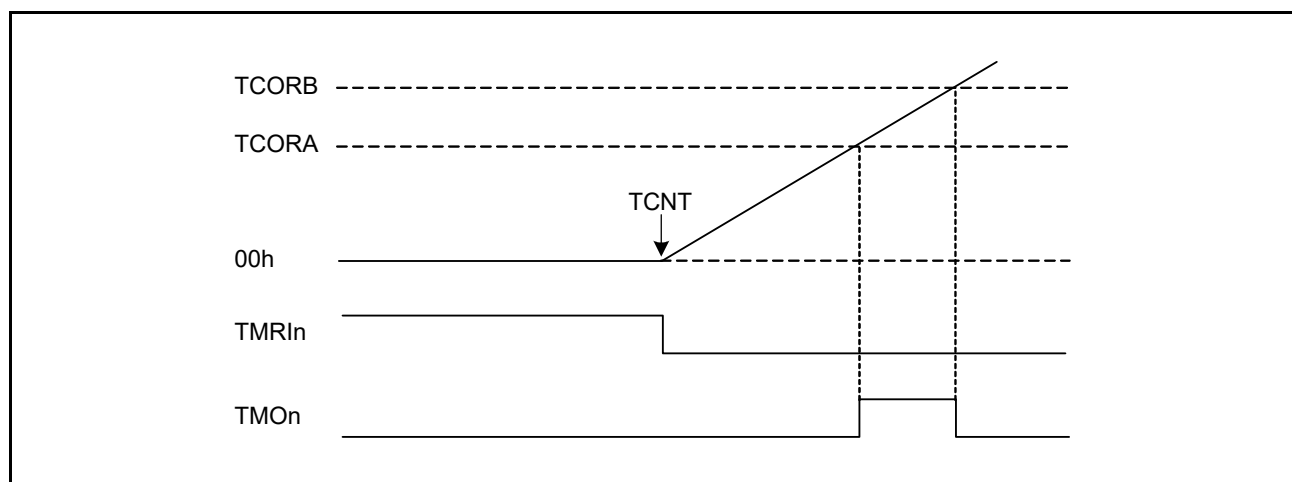


Figure 30.4 Example of External Counter Reset Signal Input (n = 0 to 3)

30.4 Operation Timing

30.4.1 TCNT Count Timing

Figure 30.5 shows the count timing of TCNT for internal clock. Figure 30.6 shows the count timing of TCNT for external clock.

Note that the external clock pulse width must be at least 1.5 PCLK cycles for increment at a single edge, and at least 2.5 PCLK cycles for increment at both edges. The counter will not increment correctly if the pulse width is less than these values.

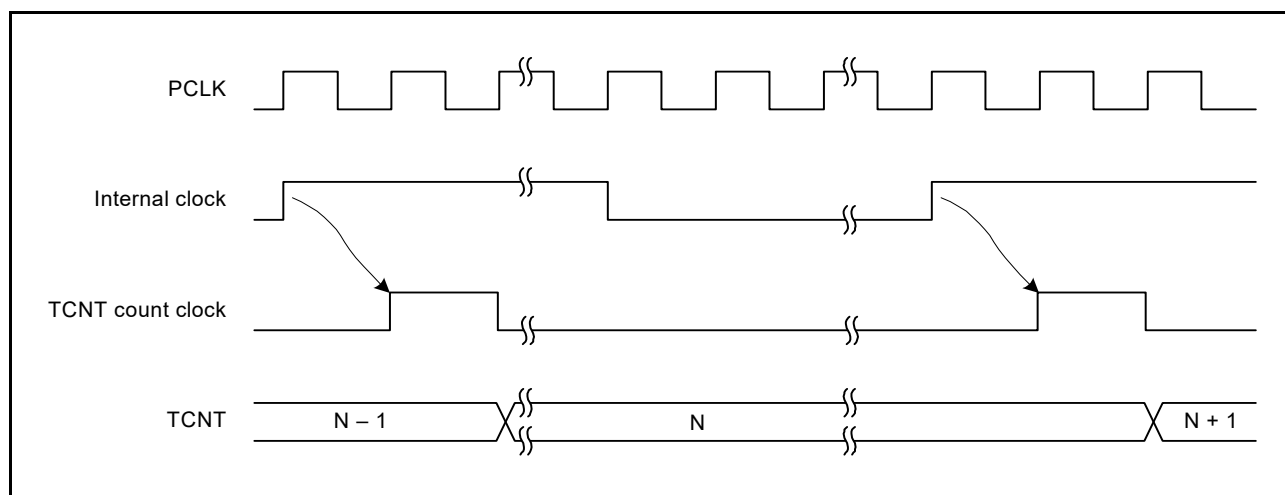


Figure 30.5 Count Timing for Internal Clock

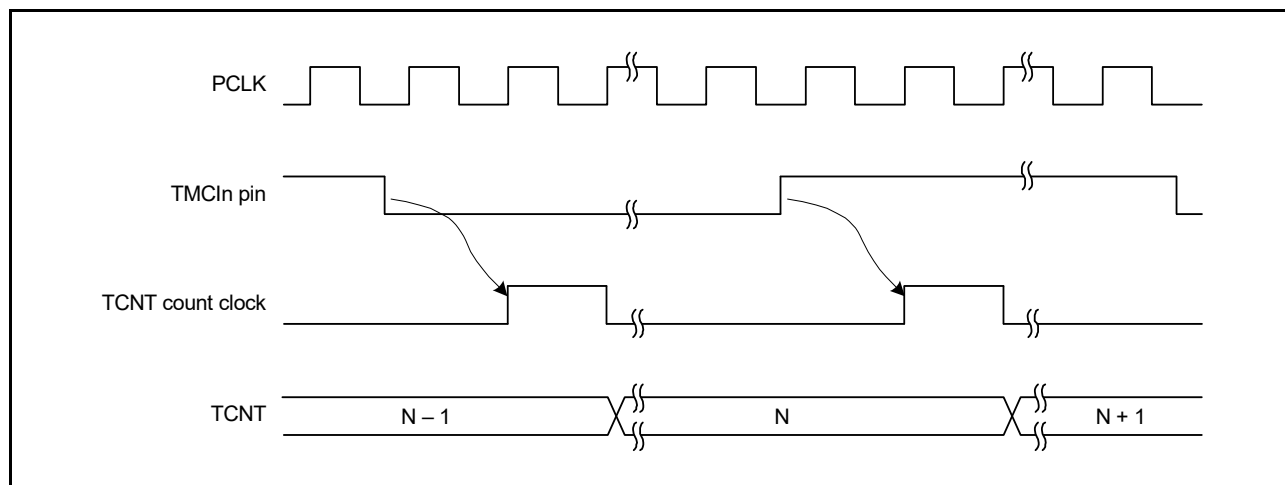


Figure 30.6 Count Timing for External Clock (at Both Edges)

30.4.2 Timing of Interrupt Signal Output on a Compare Match

A compare match refers to a match between the value of the TCORA or TCORB register and the TCNT, and a compare match interrupt signal is output at this time if the interrupt request is enabled. The compare match is generated in the last cycle in which the values match (at the time at which the value counted by TCNT to produce the match is updated). Accordingly, after a match between TCNT and the TCORA or TCORB register is detected, the compare match is not actually generated until the next cycle of the TCNT count clock. Figure 30.7 shows the timing of output of the interrupt signal.

For the corresponding interrupt vector number, refer to section 15, Interrupt Controller (ICUD) and Table 30.6.

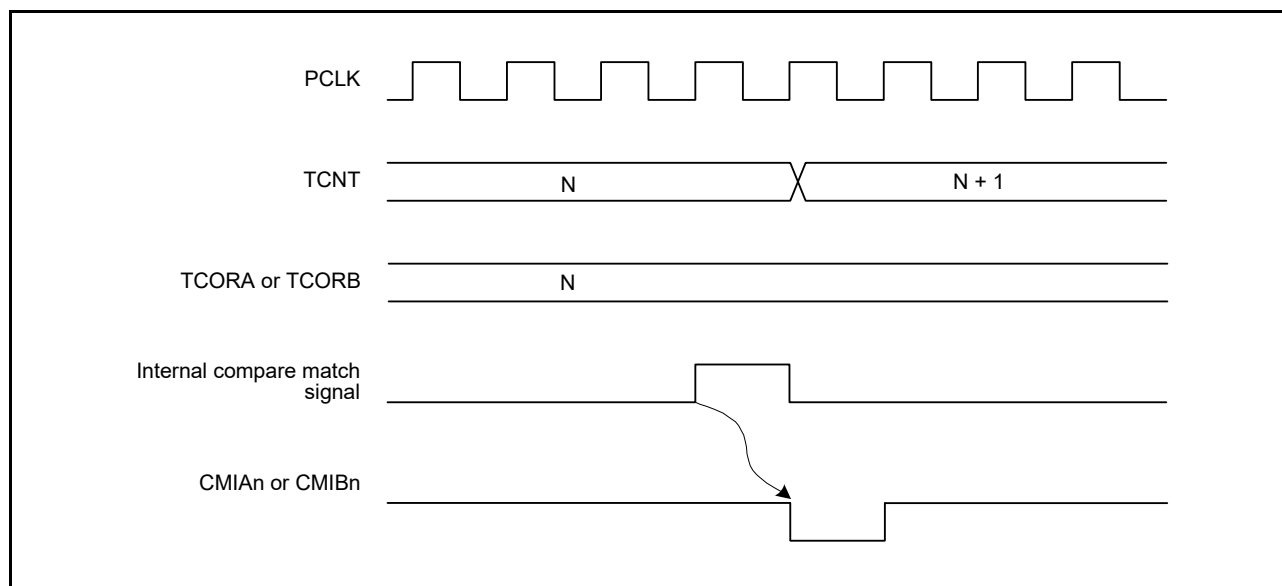


Figure 30.7 Timing of Interrupt Flag Setting to 1 at Compare Match ($n = 0$ to 3)

30.4.3 Timing of Timer Output Signal at Compare Match

When a compare match signal is generated, the output value specified by the TCSR.OSA[1:0] and OSB[1:0] bits is output on the timer output pin (TMO_n).

Figure 30.8 shows the timing when the timer output is toggled by the compare match A signal.

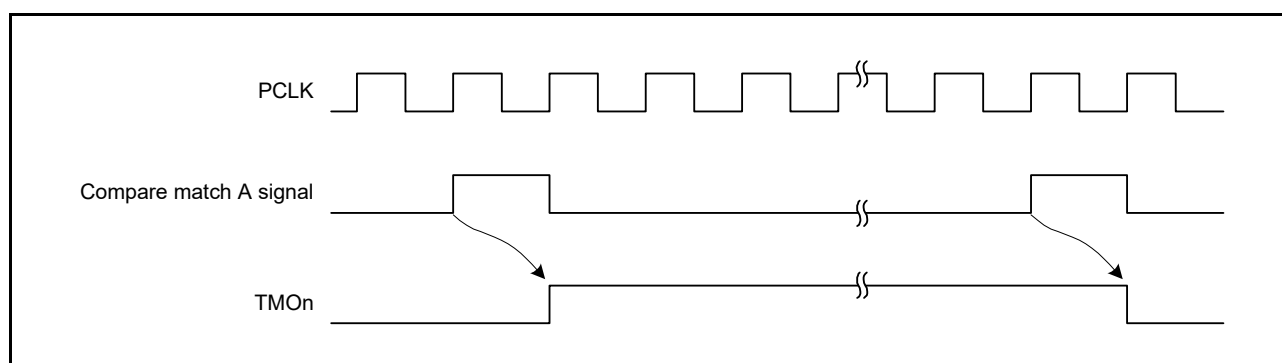


Figure 30.8 Timing of Timer Output Signal at Compare Match A Signal ($n = 0$ to 3)

30.4.4 Timing of Counter Clear by Compare Match

TCNT is cleared when compare match A or B occurs, depending on the settings of the TCR.CCLR[1:0] bits. Figure 30.9 shows the timing of this operation.

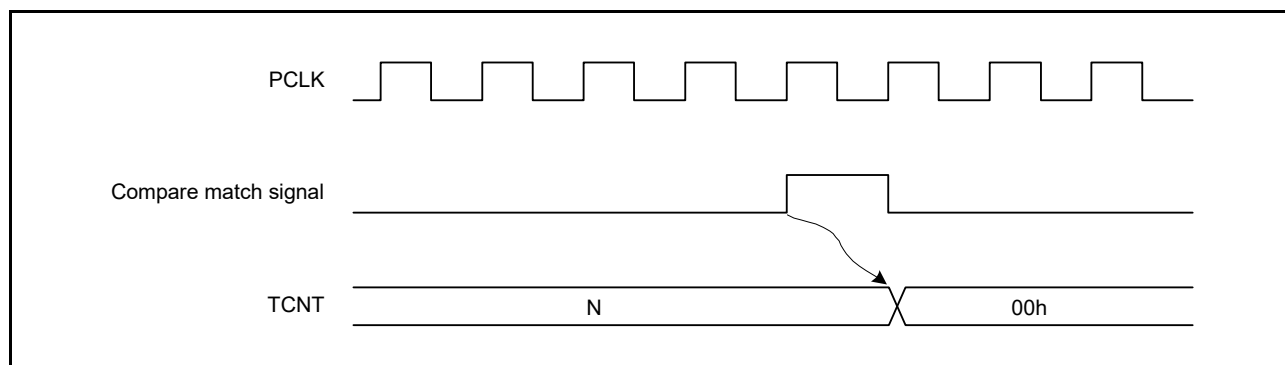


Figure 30.9 Timing of Counter Clear by Compare Match

30.4.5 Timing of the External Reset for TCNT

TCNT is cleared at the rising edge or high level of an external counter reset signal, depending on the settings of the TCR.CCLR[1:0] bits. At least 2 PCLK cycles are required from a reset input to clearing of TCNT. Figure 30.10 and Figure 30.11 show the timing of this operation.

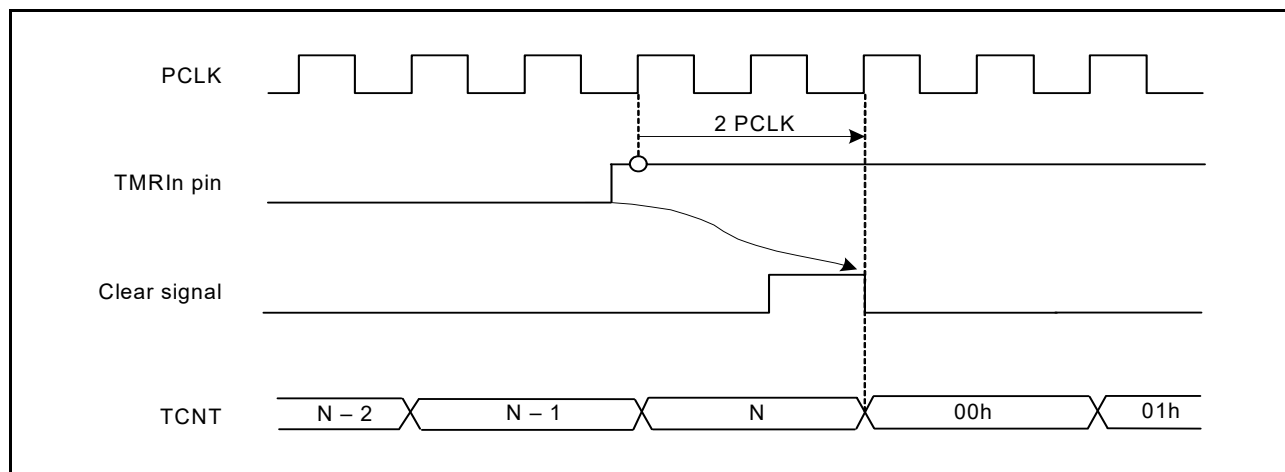


Figure 30.10 Clear Timing by External Counter Reset Signal (Rising Edge)

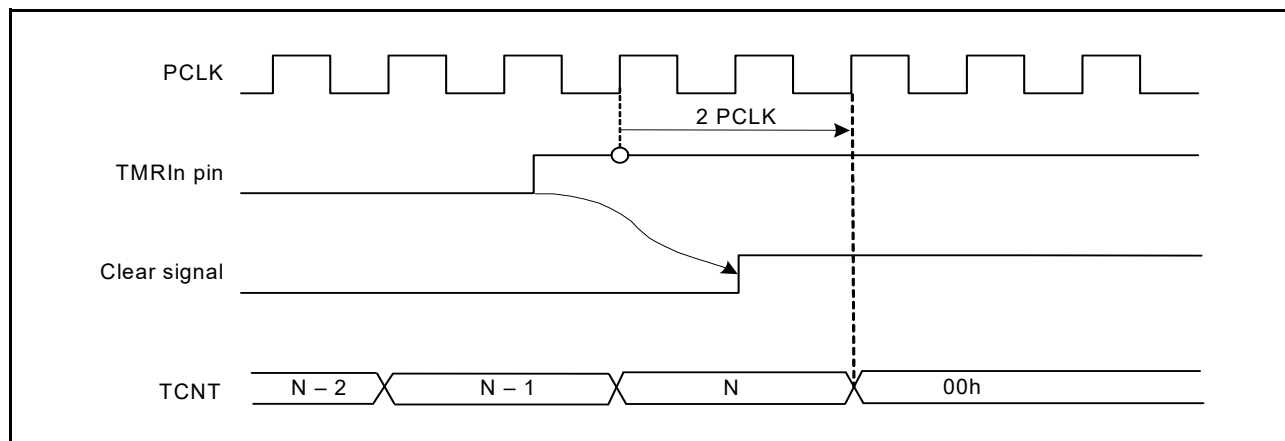


Figure 30.11 Clear Timing by External Counter Reset Signal (High Level)

30.4.6 Timing of Interrupt Signal Output on an Overflow

When TCNT overflows (changes from FFh to 00h), an overflow interrupt signal is output if this interrupt request is enabled.

Figure 30.12 shows the timing of output of the interrupt signal.

For the corresponding interrupt vector number, refer to section 15, Interrupt Controller (ICUD) and Table 30.6.

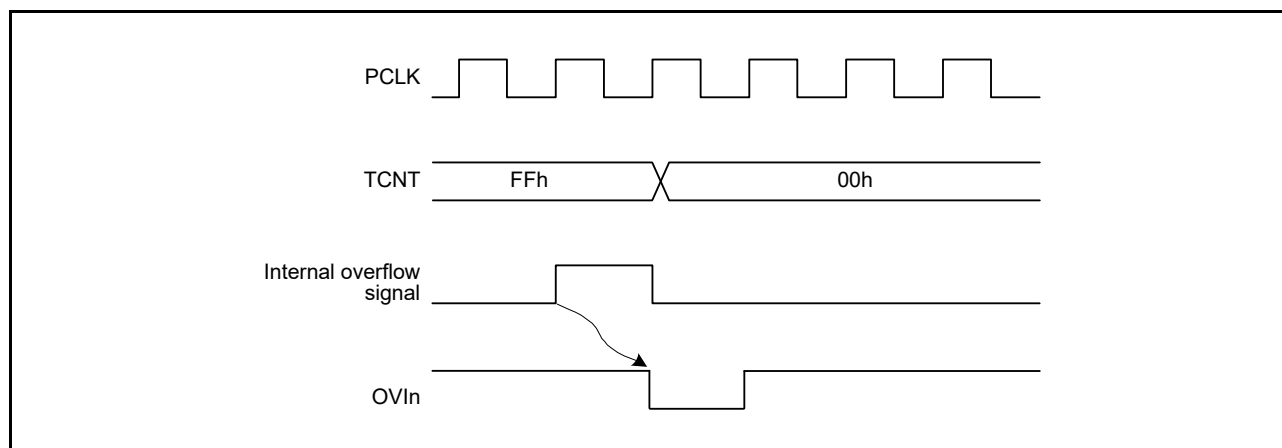


Figure 30.12 Timing of Overflow Interrupt Flag Setting to 1 (n = 0 to 3)

30.5 Operation with Cascaded Connection

If the CSS[1:0] bits in either TMR0.TCCR or TMR1.TCCR are set to 11b, the TMR of the two channels are cascaded. With this configuration, a single 16-bit timer could be used (16-bit counter mode) or compare matches of TMR0 could be counted by TMR1 (compare match count mode).

This section describes unit 0. The operation of unit 1 with cascaded connection is the same as unit 0.

30.5.1 16-Bit Count Mode

When the TMR0.TCCR.CSS[1:0] bits are set to 11b, the timer functions as a single 16-bit timer with TMR0 occupying the upper 8 bits and TMR1 occupying the lower 8 bits.

(1) Counter Clear Specification

- The settings of the TMR0.TCR.CCLR[1:0] bits become effective for the 16-bit counter. If the TMR0.TCR.CCLR[1:0] bits have been set for counter clear at compare match, the 16-bit counter (TMR0.TCNT and TMR1.TCNT together) is cleared when a 16-bit compare match event occurs. The 16-bit counter (TMR0.TCNT and TMR1.TCNT together) is cleared even if counter clear by the TMRI0 pin has been set.
- The settings of the TMR1.TCR.CCLR[1:0] bits are ignored.

(2) Pin Output

- Control of output from the TMO0 pin by the TMR0.TCSR.OSA[1:0] and OSB[1:0] bits is in accordance with the 16-bit compare match conditions.
- Control of output from the TMO1 pin by the TMR1.TCSR.OSA[1:0] and OSB[1:0] bits is in accordance with the lower 8-bit compare match conditions.

30.5.2 Compare Match Count Mode

When the TMR1.TCCR.CSS[1:0] bits are set to 11b, TMR1.TCNT counts the number of occurrences of compare match A for TMR0. TMR0 and TMR1 are controlled independently. Conditions such as generation of interrupts, output from the TMO_n pin (n = 0, 1), and counter clear are in accordance with the settings for each channel.

30.6 Interrupt Sources

30.6.1 Interrupt Sources and DTC Activation

There are three interrupt sources for TMRn: CMIA_n, CMIB_n, and OVIn. Their interrupt sources and priorities are listed in Table 30.6.

It is also possible to activate the DTC by means of CMIA_n and CMIB_n interrupts.

Table 30.6 TMR Interrupt Sources

Name	Interrupt Sources	DTC Activation
CMIA0	TMR0.TCORA compare match	Possible
CMIB0	TMR0.TCORB compare match	Possible
OVI0	TMR0.TCNT overflow	Not possible
CMIA1	TMR1.TCORA compare match	Possible
CMIB1	TMR1.TCORB compare match	Possible
OVI1	TMR1.TCNT overflow	Not possible
CMIA2	TMR2.TCORA compare match	Possible
CMIB2	TMR2.TCORB compare match	Possible
OVI2	TMR2.TCNT overflow	Not possible
CMIA3	TMR3.TCORA compare match	Possible
CMIB3	TMR3.TCORB compare match	Possible
OVI3	TMR3.TCNT overflow	Not possible

30.6.2 Startup of the A/D Converter

The compare match A of TMR0 and TMR2 allows the A/D converter to be started.

An A/D conversion start request is issued to the A/D converter in response to a generation of compare match A when the TMR_n.TCSR.ADTE bit is 1 (i.e., when an A/D conversion request in response to compare match A is enabled). In this case, the conversion trigger for the 8-bit timer should be selected in the A/D converter to start A/D conversion.

Table 30.7 Startup of A/D Converter

A/D Converter	TMR Unit No.	Target	A/D Conversion Start Request
S12AD, S12AD1 (12-bit A/D converter)	0	Compare match between TMR0.TCORA and TMR0.TCNT	TMTRG0AN_0
	1	Compare match between TMR2.TCORA and TMR2.TCNT	TMTRG0AN_1

30.7 Link Operation by ELC

30.7.1 Event Signal Output to ELC

The TMR uses the event link controller (ELC) to perform link operation to the previously specified module using the interrupt request signal as the event signal. The TMR outputs compare match A, compare match B, and overflow signals as event signals. Channels that can be used in this way are TMR0 to TMR3.

The event signal can be output regardless of the setting of the corresponding interrupt request enable bits (TMRn.TCR.OVIE, TMRn.TCR.CMIEA, and TMRn.TCR.CMIEB (n = 0 to 3)). For details, refer to **section 21, Event Link Controller (ELC)**.

The event output function can be used for the cascaded operation.

30.7.2 TMR Operation when Receiving an Event Signal from ELC

The TMR can perform either of the following operations upon the event previously specified by the ELSRn register of the ELC. However, the ELC does not support the cascaded operation.

(1) Count Start

When the TMR count start operation is selected by the ELOPD register of the ELC and the event specified by ELSRn occurs, the TCSTR.TCS bit is set to 1, starting the TMR count operation. After the TMR count start operation is selected by the ELOPD register of the ELC, use the TCCR.CKS[2:0] and CSS[1:0] bits to select the count source.

If the specified event occurs while the TCS bit is 1, the event is ignored.

Write 0 to the TCSTR.TCS bit to stop counting.

When the count start event is input in the count stopped state, the TMR starts counting again according to the CKS[2:0] and CSS[1:0] bits.

The TCS bit is valid only when the ELOPD.TMR0MD[1:0], ELOPD.TMR1MD[1:0], ELOPD.TMR2MD[1:0], and ELOPD.TMR3MD[1:0] bits of the ELC select the count start operation.

(2) Event Count

When the TMR event count operation is selected by the ELOPD register of the ELC and the event specified by ELSRn occurs, the events are counted as the count source regardless of the TCCR.CKS[2:0] and CSS[1:0] bit settings. Reading the counter value returns the number of events that have been actually input.

(3) Count Restart

When the TMR count restart operation is selected by the ELOPD register of the ELC and the event specified by ELSRn occurs, the TCNT counter value is modified to the initial value. If the CKS[2:0] and CSS[1:0] bit settings are not disabling the clock input, the count operation is continued.

30.7.3 Notes on Operating TMR According to an Event Signal from ELC

The following describes the notes on operating the TMR using the event link feature.

(1) Count Start

When the event specified by `ELSRn` occurs during the write cycle to the `TCSTR.TCS` bit, the cycle is not completed; setting 1 according to the event occurrence takes priority.

(2) Event Count

When the event specified by `ELSRn` occurs during the write cycle to the `TCNT`, the cycle is not completed; event count operation according to the event occurrence takes priority.

(3) Count Restart

When the event specified by `ELSRn` occurs during the write cycle to the `TCNT`, the cycle is not completed; count value initialization according to the event occurrence takes priority.

30.8 Usage Notes

30.8.1 Module Stop State Setting

Operation of the TMR can be disabled or enabled by using the module stop control registers. The initial setting is for halting of TMR operation. Register access becomes possible after release from the module stop state. For details, refer to section 11, Low Power Consumption.

30.8.2 Notes on Setting Cycle

If the compare match is selected for counter clear, TCNT is cleared at the last PCLK in the cycle in which the value of TCNT matches with that of TCORA or TCORB. TCNT updates the counter value at this last state. Therefore, the counter frequency is obtained by the following formula (f: Counter frequency, PCLK: Operating frequency, N: TCORA and TCORB register setting value).

$$f = \text{PCLK} / (N + 1)$$

30.8.3 Conflict between TCNT Write and Counter Clear

If a counter clear signal is generated concurrently with CPU write to TCNT, the clear takes priority and the write is not performed as shown in Figure 30.13.

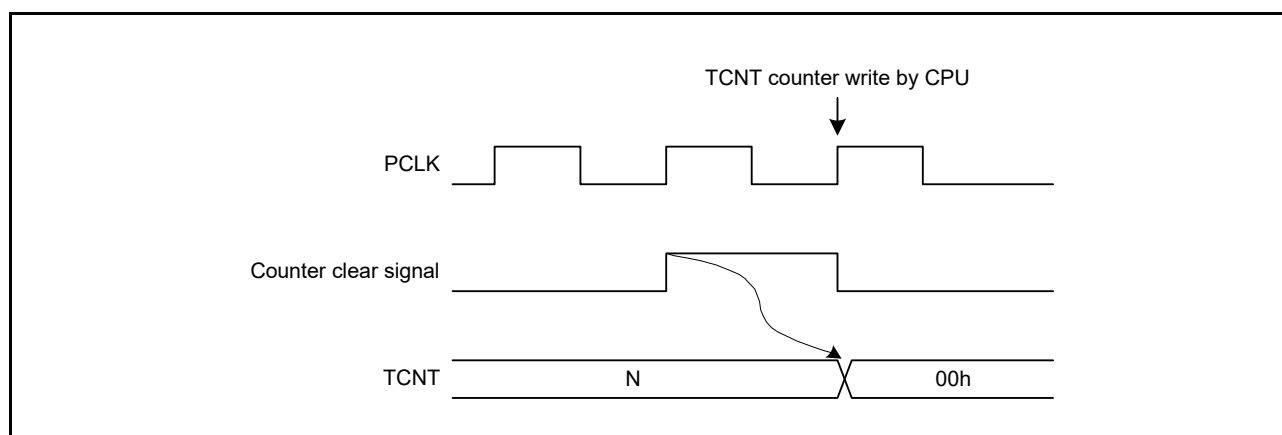


Figure 30.13 Conflict between TCNT Write and Counter Clear

30.8.4 Conflict between TCNT Write and Increment

Even if a counting-up signal is generated concurrently with CPU write to TCNT, the counting-up is not performed and the write takes priority as shown in Figure 30.14.

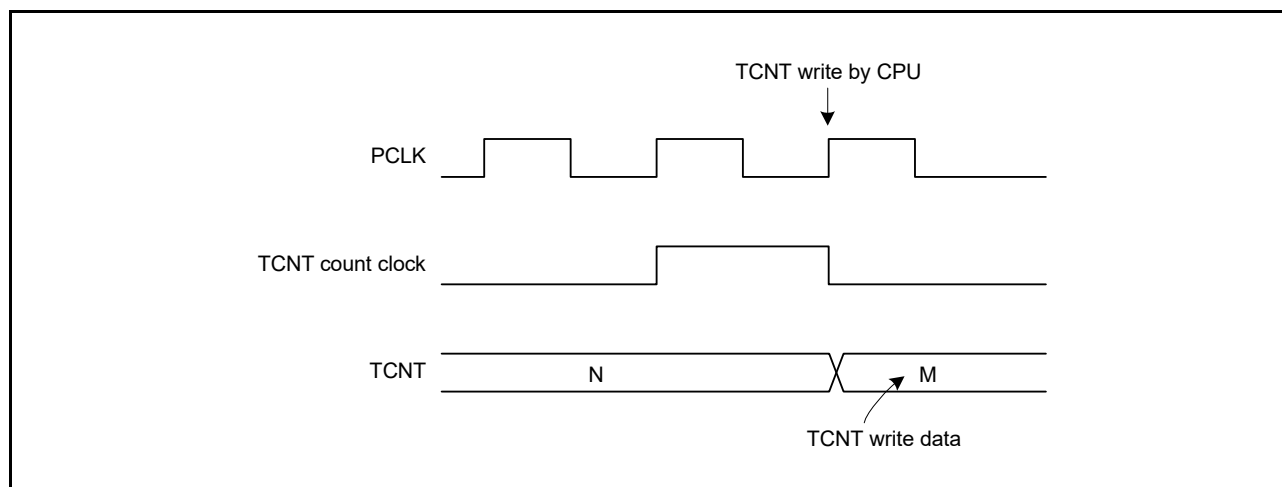


Figure 30.14 Conflict between TCNT Write and Increment

30.8.5 Conflict between TCORA or TCORB Write and Compare Match

Even if a compare match signal is generated simultaneously with CPU write to TCORA or TCORB as shown in Figure 30.15, the write takes priority and the compare match signal does not reach High level.

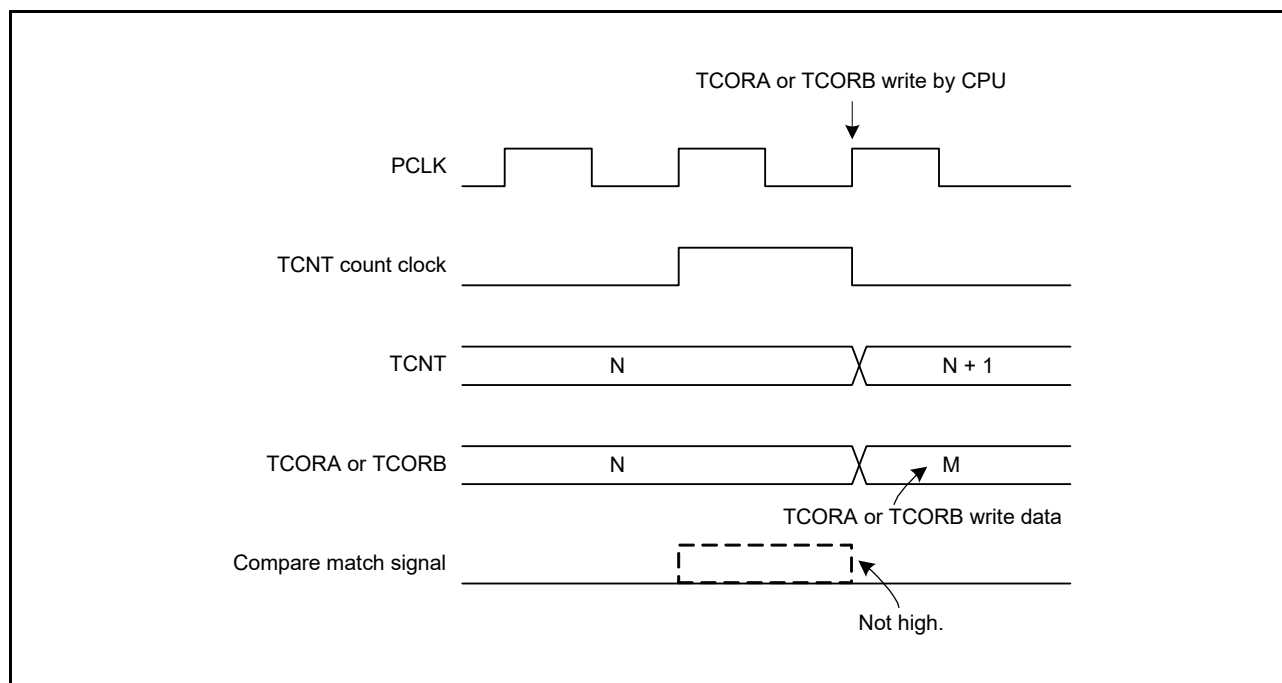


Figure 30.15 Conflict between TCORA or TCORB Write and Compare Match

30.8.6 Conflict between Compare Matches A and B

If compare match events A and B occur at the same time, the 8-bit timer operates in accordance with the priorities for the output methods high for compare match A and compare match B, as listed in Table 30.8.

Table 30.8 Timer Output Priorities

Output Setting	Priority
Toggle output	High
High output	↑
Low output	↑
No change	Low

30.8.7 Switching of Internal Clocks and TCNT Operation

TCNT may be incremented erroneously depending on when the internal clock is switched. Table 30.9 lists the relationship between the timing at which the internal clock is switched (by writing to the TCCR.CKS[2:0] bits) and the operation of TCNT.

When TCNT count clock is generated from an internal clock, the rising edge of the internal clock pulse are always monitored. If the signal levels of the clocks before and after switching change from low to high as shown in No. 2 in Table 30.9, the change is considered as an edge. Therefore, a TCNT count clock is generated and TCNT is incremented. The erroneous increment of TCNT can also happen when switching between internal and internal clocks.

Table 30.9 Switching of Internal Clocks and TCNT Operation (1/2)

No.	Timing to Change the TCCR.CKS[2:0] Bits	TCNT Counter Operation
1	Switching from low to low*1	TCNT TCCR.CKS[2:0] bits changed
2	Switching from low to high*2	TCNT TCCR.CKS[2:0] bits changed

Table 30.9 Switching of Internal Clocks and TCNT Operation (2/2)

No.	Timing to Change the TCCR.CKS[2:0] Bits	TCNT Counter Operation
3	Switching from high to low*4	TCCR.CKS[2:0] bits changed
4	Switching from high to high	TCCR.CKS[2:0] bits changed

Note 1. Includes switching from low to stop, and from stop to low.

Note 2. Includes switching from stop to high.

Note 3. Generated because the change of the signal levels is considered as an edge; TCNT counter is incremented.

Note 4. Includes switching from high to stop.

30.8.8 Clock Source Setting with Cascaded Connection

If 16-bit counter mode and compare match count mode are specified at the same time, count clocks for TMR0.TCNT and TMR1.TCNT (TMR2.TCNT and TMR3.TCNT) are not generated, and the counter stops. Do not specify 16-bit counter mode and compare match count mode simultaneously.

30.8.9 Continuous Output of Compare Match Interrupt Signal

When TCORA or TCORB is set to 00h, PCLK/1 is set as the internal clock, and compare match is set as the counter clear source, the TCNT counter remains 00h and is not updated, and a compare match interrupt signal is output continuously to form a flat signal level.

At this time, the interrupt controller cannot detect the second and subsequent interrupts.

Figure 30.16 shows operation timing when the compare match interrupt signal is continuously output.

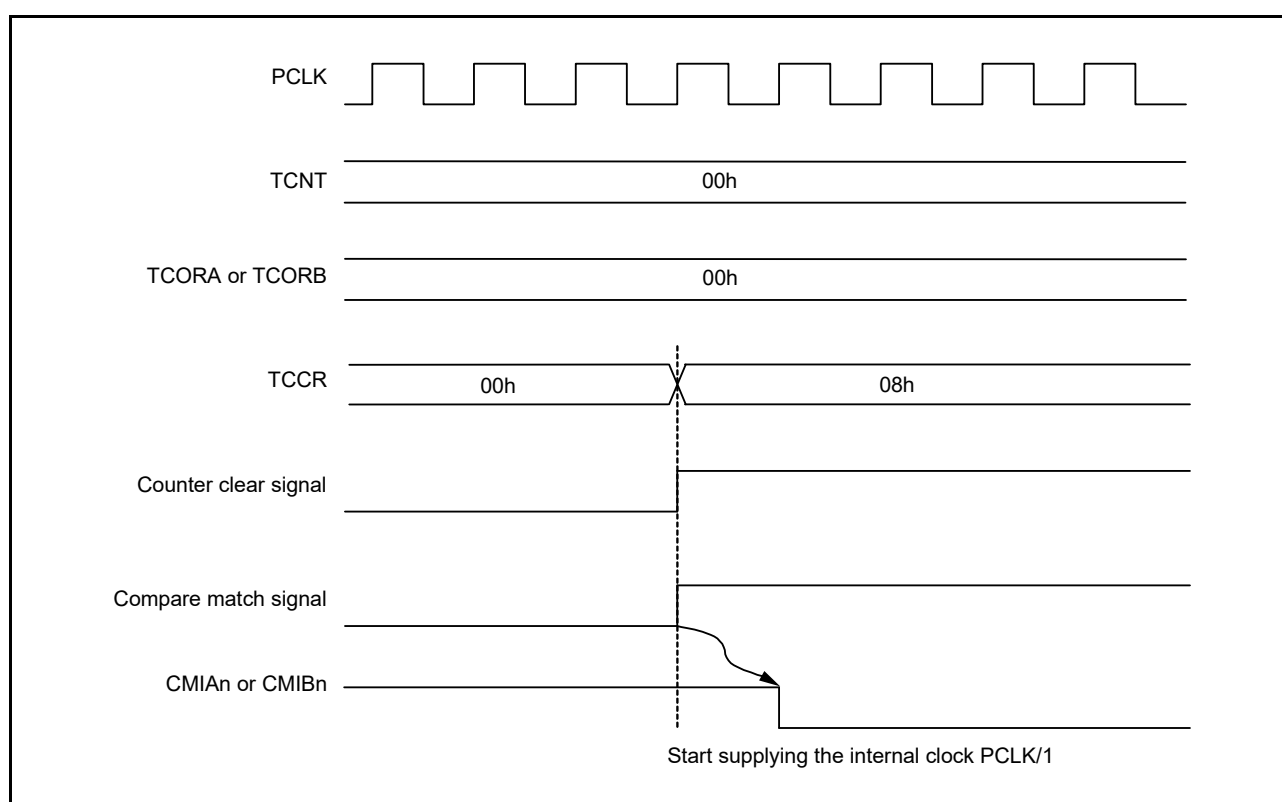


Figure 30.16 Continuous Output of Compare Match Interrupt Signal (n = 0 to 3)